

Pranav Maheshwari

AI & Machine Learning Engineer | Physics-Informed AI & LLMs

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Status: EU Working Student Visa (Full Work Eligibility in Germany)

[linkedin.com/in/pranavmaheshwari0617](https://www.linkedin.com/in/pranavmaheshwari0617) | github.com/pranavmaheshwari

PROFESSIONAL SUMMARY

Machine Learning & AI Engineer pursuing M.Sc. in AI Engineering at Universität Passau, currently contributing to PhysicsNeMo surrogate modeling pipelines at Mercedes-Benz AG. Experienced in building physics-informed neural networks (PINNs), LLM applications, CUDA-accelerated deep learning pipelines, and production MLOps data workflows using PyTorch, SQL, and Zarr. Adept at translating complex engineering simulations into fast, scalable AI models.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript (React.js), Bash

AI / ML Frameworks: PyTorch, PhysicsNeMo, PINNs (Physics-Informed Neural Networks), OpenCV, YOLOv3/YOLOv8, LLMs & Transformers, CUDA Acceleration, Scikit-Learn

Data Engineering Tools: Zarr Data Format, Mesh Processing, Tableau, Git, REST APIs, Docker, PostgreSQL

PROFESSIONAL EXPERIENCE

Working Student — Physics-Informed AI | Mercedes-Benz AG

08/2025 – Present

Sindelfingen, Germany

- Researched and benchmarked PhysicsNeMo surrogate AI models across complex engineering use cases, evaluating linear elasticity models for real-time stress/displacement field predictions.
- Engineered end-to-end data training pipelines, converting simulation data into PhysicsNeMo-trainable datasets using mesh processing, geometry-field alignment, and Zarr structured storage.
- Automated physics data preprocessing and Curator-style workflows, accelerating surrogate model iteration speed for engineering design optimization.

Market Research & Forecasting Intern | EventCart

12/2022 – 03/2023

Noida, India

- Architected SQL-based data workflows to preprocess 100K+ market and customer records for forecasting models.
- Developed predictive machine learning models to identify key market trends and consumer behavior patterns, boosting forecast precision.
- Designed interactive Tableau reporting dashboards to visualize model metrics and business insights for executive teams.

Product Engineer | Highradius

01/2022 – 04/2022

Delhi, India

- Deployed an AI-enabled B2B invoicing platform with automated invoice processing pipelines, optimizing data extraction accuracy.
- Integrated RESTful APIs to automate invoice validation and decision-making workflows, reducing manual processing time by 40%.
- Built a responsive React.js web dashboard featuring real-time data editing, advanced search, and PDF generation tools.

KEY TECHNICAL PROJECTS

Real-Time Vehicle Collision Detection System | github.com/pranavmaheshwari/collision-detection

PyTorch, YOLOv3, OpenCV, CUDA, Python, REST APIs

- Engineered a real-time object tracking and collision detection system using YOLOv3 and OpenCV with CUDA GPU acceleration.
- Implemented automated instant SMS/email notification APIs for emergency alerts, reducing detection-to-alert latency under 200ms.
- Authored research paper detailing real-time multi-vehicle tracking architecture and GPU memory optimization benchmarks.

EDUCATION

Universität Passau | Master of Science in Artificial Intelligence Engineering

10/2023 – Present

Passau, Germany • Focus on Deep Learning Architectures, Physics-Informed Neural Networks (PINNs), LLMs, and Advanced Pipelines.

SRM Institute of Science and Technology | B.Tech in Computer Science & Engineering

07/2019 – 05/2023

India • Graduated with Computer Science specialization. Coursework: Data Structures, Algorithms, AI, Database Systems.

CERTIFICATIONS & LANGUAGES

- **Google Data Analytics Professional Certificate** (Coursera / Google, 2023)
- **The Complete Python Pro Bootcamp** (Udemy, 2023)
- **Languages:** English (Full Professional / C1), German (Working Knowledge / A2-B1)